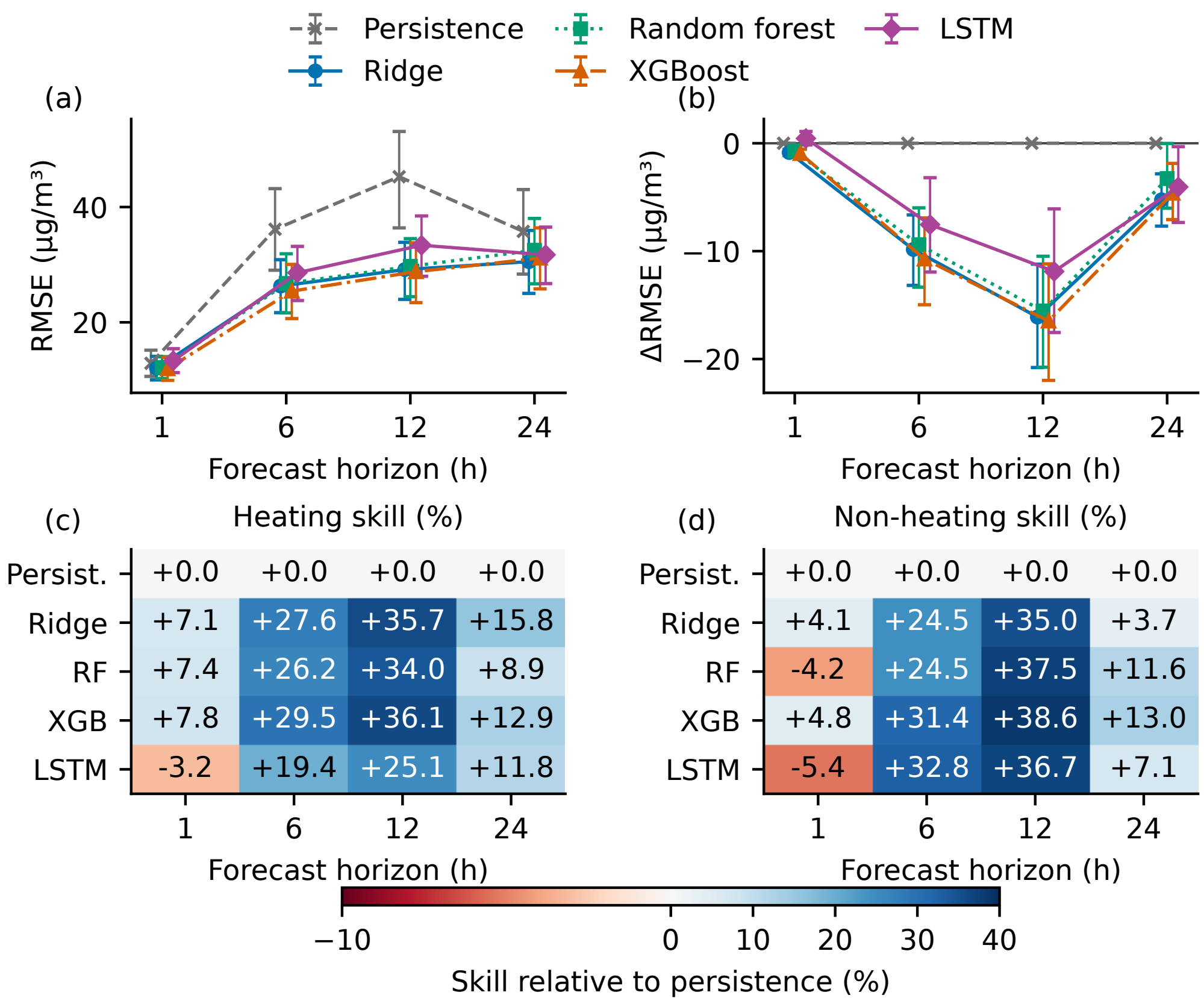
Persistence
Random forest
LSTM
Ridge
XGBoost
(a)
RMSE (µg/m³)
40
20
1
6
12
24
Forecast horizon (h)
(b)
ΔRMSE (µg/m³)
0
−10
−20
1
6
12
24
Forecast horizon (h)
(c)
Heating skill (%)
Persist. +0.0 +0.0 +0.0 +0.0
Ridge +7.1 +27.6 +35.7 +15.8
RF +7.4 +26.2 +34.0 +8.9
XGB +7.8 +29.5 +36.1 +12.9
LSTM -3.2 +19.4 +25.1 +11.8
1
6
12
24
Forecast horizon (h)
(d)
Non-heating skill (%)
Persist. +0.0 +0.0 +0.0 +0.0
Ridge +4.1 +24.5 +35.0 +3.7
RF -4.2 +24.5 +37.5 +11.6
XGB +4.8 +31.4 +38.6 +13.0
LSTM -5.4 +32.8 +36.7 +7.1
1
6
12
24
Forecast horizon (h)
−10
0
10
20
30
40
Skill relative to persistence (%)